

Predicting Employee Attrition with Tree-Based Machine Learning Models: A Case Study at Salifort Motors

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Abstract—Employee attrition poses a critical challenge for organizations seeking to retain talent and maintain productivity. This study investigates employee turnover at Salifort Motors using a dataset containing behavioral, demographic, and performance-related variables. Multiple modeling approaches, including logistic regression, decision trees, and random forests, were developed and compared. After preprocessing and feature engineering, models were trained, validated, and evaluated on a held-out test set. Logistic regression achieved an *accuracy* of 83% but was outperformed by tree-based models. The optimized random forest achieved an *AUC* of 0.96 and *accuracy* of 96.2%, with feature importances highlighting last evaluation, number of projects, tenure, and workload as the strongest predictors of attrition. Feature engineering further confirmed that employee overwork and mid-tenure dissatisfaction were key drivers of turnover. The findings suggest that actionable policies—such as capping project loads, adjusting promotion cycles, and clarifying workload expectations—could improve employee retention. This case study demonstrates the value of predictive analytics in human resources and underscores the trade-off between model performance and interpretability.

Index Terms—Employee Attrition, HR Analytics, Machine Learning, Logistic Regression, Decision Tree, Random Forest, Predictive Modeling, Employee Retention.

I. INTRODUCTION

Employee retention is a persistent concern in organizational management, particularly in competitive industries where turnover imposes financial, cultural, and operational costs. Predicting attrition has therefore become a central task for human resources (HR) analytics. Recent advances in machine learning provide the opportunity to model complex, nonlinear relationships between employee attributes and turnover, enabling actionable insights for policy and management interventions.

This project presents a case study of *SalifortMotors*, a large manufacturing company facing elevated employee turnover. Using company HR records, we develop and evaluate predictive models to identify factors associated with attrition and to assess which employees are at risk of leaving. Specifically, we compare classical statistical modeling (logistic regression) with modern machine learning approaches (decision trees and random forests). The objectives are threefold:

To quantify the predictive accuracy of alternative modeling approaches.

To identify the most influential drivers of employee attrition.

To translate model outputs into actionable retention strategies for Salifort Motors.

II. DATA AND METHODOLOGY

A. Dataset Overview

The dataset comprises approximately 15,000 employee records with features covering satisfaction level, last evaluation score, average monthly hours, number of projects, tenure (years at the company), recent promotions, department, and salary level. The binary target variable indicates whether an employee left the company.

B. Preprocessing & Feature Engineering

- **Data cleaning:** Ensured no missing values and standardized categorical variables.
- **Feature engineering:** Created a binary “overworked” variable (1 if monthly hours > 175, else 0) to capture workload stress. Dropped potentially leaky features such as satisfaction level for robustness testing.
- **Train–validate–test split:** Data partitioned into 70% training, 15% validation, and 15% testing.
- **Normalization:** Standardization applied to continuous features for logistic regression; tree-based models did not require scaling.

C. Modeling Approaches

- **Baseline Logistic Regression:** Provides interpretable coefficients to establish a benchmark.
- **Decision Tree Classifier:** Tuned using cross-validated grid search across depth, split, and leaf parameters.
- **Random Forest Classifier:** Ensemble of trees, optimized over depth, number of estimators, and sample ratios. Regularization was applied to reduce overfitting.

Performance was assessed using *accuracy*, *precision*, *recall*, *F1 – score*, and area under the *ROC curve (AUC)*.

III. EXPLORATORY DATA ANALYSIS

EDA revealed several intuitive patterns:

- Employees with very low or very high evaluation scores were more likely to leave.
- Overworked employees (175 monthly hours) displayed elevated attrition risk.
- Tenure effects: Attrition peaked around four years of employment, suggesting a mid-career dissatisfaction phenomenon.
- Salary: Employees with low salary levels had a substantially higher probability of leaving.

Fig 1



Fig. 1: Scatterplot of satisfaction vs. monthly hours by attrition status (orange = left, blue = stayed). Two high-risk clusters appear: moderate hours with mid satisfaction, and very high hours with low satisfaction.

IV. RESULTS AND INTERPRETATION

A. Logistic Regression Baseline

TABLE I: Performance of Logistic Regression (Baseline)

Metric	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	0.83	0.80	0.83	0.80	0.83

Logistic regression revealed statistically significant coefficients for evaluation score, number of projects, and salary. However, its linear assumptions limited its ability to capture interactions between features. Fig 2

B. Decision Tree (Round 1)

TABLE II: Performance of Decision Tree (Round 1)

Metric	Accuracy	Precision	Recall	F1-score	AUC
Decision Tree	0.972	0.95	0.96	0.95	0.97

Interpretation: Decision trees offered strong predictive accuracy, with interpretability via feature importance rankings. Key splits occurred on last evaluation, number of projects, and tenure.

C. Random Forest (Round 1)

Random forest outperformed the decision tree across all metrics except recall, which was only marginally lower. Importantly, the ensemble reduced overfitting risk.

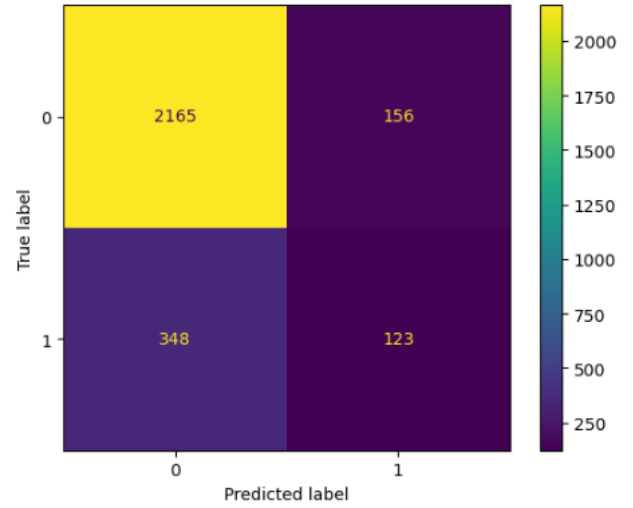


Fig. 2: Confusion matrix for the logistic regression model. The model captures general attrition patterns but shows moderate misclassification.

TABLE III: Performance of Random Forest (Round 1)

Metric	Accuracy	Precision	Recall	F1-score	AUC
R F (Round 1)	0.981	0.96	0.97	0.96	0.98

D. Feature Engineering & Round 2 Models

TABLE IV: Performance of Random Forest (Round 2)

Metric	Accuracy	Precision	Recall	F1-score	AUC
R F (Round 2)	0.962	0.87	0.90	0.89	0.94

This indicates stable, generalizable performance on unseen data.

E. Feature Importance Analysis

Both tree-based models highlighted four critical predictors:

- Last evaluation
- Number of projects
- Tenure
- Overworked status

(Insert: Figure 3. Random Forest Feature Importances)

Fig 3

Fig 4

Interpretation: Attrition is driven by workload intensity, lack of advancement after four years, and imbalances in evaluation/reward systems.

V. LIMITATIONS

- **Data leakage:** Certain features (e.g., satisfaction level) may not be observable in practice for all employees.
- **Temporal context:** The dataset lacks time-series variation; attrition may vary across seasons or economic cycles.
- **Generalizability:** Data is company-specific; findings may not apply to different organizational cultures.

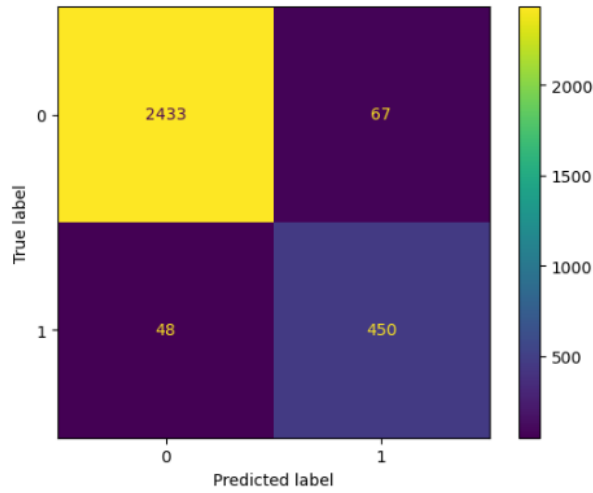


Fig. 3: Confusion matrix for the improved decision tree model. Random forest slightly outperformed the decision tree, confirming robustness.

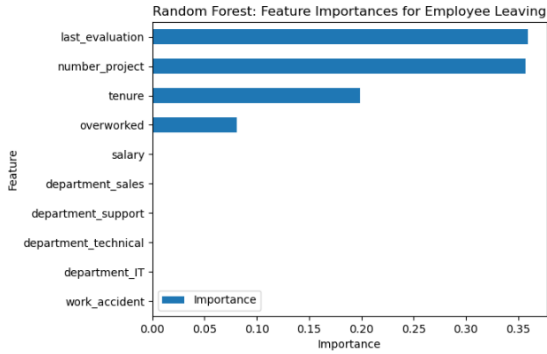


Fig. 4: Feature importance plot from the random forest model. *Last_evaluation*, *number_project*, *tenure*, and *overworked* are the strongest predictors of employee attrition.

- **Fairness & ethics:** Predictive models risk stigmatizing employees flagged as “at-risk,” raising concerns around bias and responsible use.

VI. CONCLUSION AND FUTURE WORK

This study demonstrates the utility of machine learning models in predicting employee attrition at *SalifortMotors*. Logistic regression provided interpretable but limited baselines, while tree-based models—especially random forests—delivered superior predictive power, with AUC exceeding 0.95.

Key actionable insights include:

- Avoiding employee overload by capping project assignments.
- Addressing dissatisfaction around the four-year tenure mark through promotions or re-engagement initiatives.
- Rewarding workload fairly and clarifying overtime expectations.

Future work will incorporate temporal analysis, explore explainable AI methods (e.g., SHAP values), and validate findings against external datasets to strengthen generalizability.

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VII. GITHUB REPOSITORY

All code and data used in this study are available at: <https://github.com/RoniF-pixel/Python-Projects/tree/main/Salifort%20Motors>